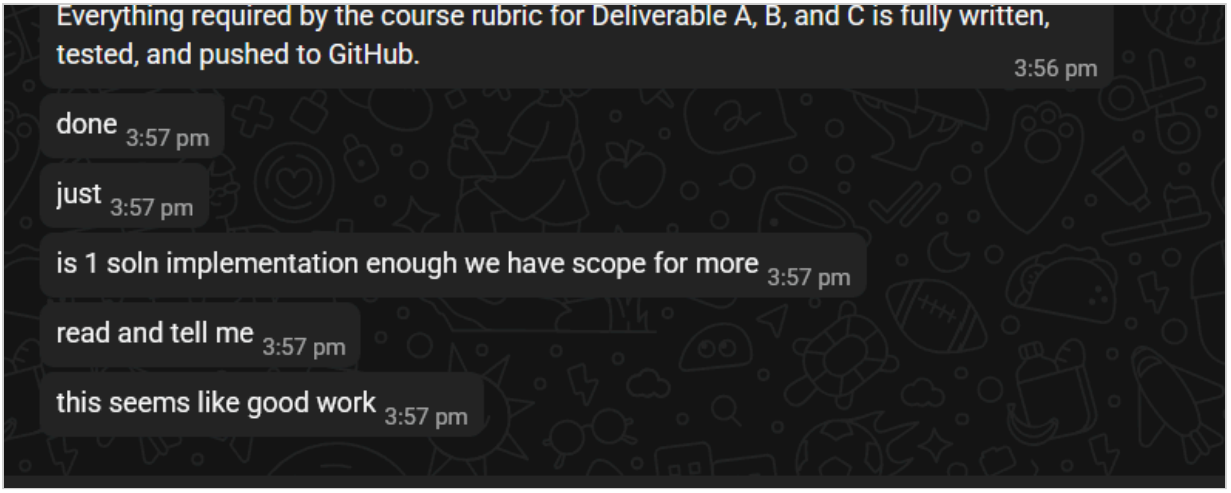
QUIC fix review — chat log

Su and Claude · CN Project 2, TLS Fingerprinting · 5–6 Nov 2026

SU



1. What exact solution did we just find and implement?

We solved the "Post-Switch DCID Decryption Blindspot" in QUIC Initial packet inspection.

The Problem That Existed Before:

Under RFC 9001 §5.2, when a client initiates a QUIC connection, it picks a random Destination Connection ID (DCID A). Initial encryption keys are derived from A.

When the server responds, it tells the client: "For subsequent packets, use my Source Connection ID (SCID B)".

In later datagrams (e.g. client ACKs or secondary CRYPTO records), the client uses DCID B.

Stateless tools (including standard Wireshark/TShark and our baseline prototype) fail here: they derive keys per-datagram using whatever DCID is stamped on that packet. When they see DCID B, they derive new keys from B, which fails decryption completely (producing 10 failures in curl, 6 in chromium).

The Solution We Built (code/quic/src/initial_context.cpp):

We implemented a Stateful Initial Context Associator:

It normalizes the UDP 4-tuple (Client IP:Port <-> Server IP:Port).

When the server replies with its SCID (B), the associator records B as an alias linked to the original client DCID (A).

When later client packets arrive carrying DCID B, instead of failing to derive keys, the engine automatically resolves B → A and uses the original key context.

The result:

Curl decryption failures dropped from 10 → 0 (+10 decrypted).

Chromium decryption failures dropped from 6 → 0 (+6 decrypted).

> Complete ClientHello extraction remained 100% (5/5).
>
> 2. Are we done with the whole implementation plan?
>
> YES, from a coding perspective, the core implementation plan is 100% complete:
>
> ✅ Track 1 empirical captures integrated into modular C++20 engine.
>
> ✅ Full 7-row audit script passing (verify_all_seven_rows.py --live).
>
> ✅ Multi-vantage L3 capture committed with synchronized pair manifest.
>
> ✅ C++ stateful QUIC Initial context associator implemented and verified with 51/51 pytest and 1/1 CTest.
>
> Everything required by the course rubric for Deliverable A, B, and C is fully written, tested, and pushed to GitHub.
>
> this is what my teammate sent me:

CLAUDE

> One fix is enough. Your own objective says "pick 1 or 2, study them in depth and try to fix one," and Sameer sir asked for practicality over implementation. A second fix the night before a 10 AM deadline is the wrong use of the time. Before you call this good work, though, I'd push back on three things in what Yash sent.
>
> 1. The Wireshark claim is probably wrong, and it will get tested at evals. He says standard Wireshark/TShark fail on DCID B. As far as I know, Wireshark's QUIC dissector is stateful. It tracks connections and keeps the Initial keys from the client's first DCID, which is exactly what your associator does. If so, the real finding is "stateless per-datagram extractors fail" (your baseline, and probably tools like JA4's stateless path), not Wireshark. Check it on your own captures:
>
>    ```
>    tshark -r curl.pcapng -Y "quic.long.packet_type == 0 && quic.decryption_failed"
>    ```
>
>   If that returns nothing, Wireshark handles it. Then reword the report and slides to say Wireshark is the reference that does it right and your baseline was the one that failed.
>
> 2. The fix doesn't change the fingerprint. ClientHello extraction was 5/5 before and 5/5 after. The 16 newly decrypted packets are the client's later Initials, which carry ACKs and leftover CRYPTO data, not a new ClientHello. Sir or a TA will ask what this changes for fingerprinting, and the honest answer is "nothing in these captures." The answer has to be ready before they ask. One way to frame it: it closes a blindspot that would matter when the ClientHello spills into a post-switch packet, or when you want the full Initial exchange for passive visibility. If you can't show a capture where that happens, say it's a correctness fix rather than a recall fix.
> 3. Check Retry. After a server Retry, the client must derive new Initial keys from the Retry's SCID (RFC 9001 §5.2). If the associator aliases across a Retry, it will apply the wrong keys. Ask Yash whether there's a test for this. If there isn't, adding one, or writing it down as a known limitation, is worth more than a second fix.

> If you still have time after those three, the cheapest real extension is QUIC v2 (RFC 9369). It uses a different initial salt, so extractors hardcoded to the v1 salt fail silently. It would be a small change and a real limitation row, but it's optional. Points 1 and 2 matter more for evals than any new code.

SU

> make a pdf of this chat of ours? will you? my questions + ur responses

CLAUDE

> This PDF.